Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Senior Software Engineer · LangGraph · Multi-Agent · RAG · LLMOps · Python · AWS · PropTech · 8+ Years Production SaaS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | portfolio | McLean area, Virginia — Remote

PROFESSIONAL SUMMARY

AI engineer and senior software engineer with **8+ years shipping production SaaS** at Entrata (property management platform, direct AppFolio/RealPage competitor) and **3+ years hands-on agentic AI**. Built **AgentProbe**: open-source multi-agent eval framework - 3-agent LangGraph runtime, 5-metric LLM-as-judge engine, CI/CD regression gates, React observability dashboard. Built **Leasel**: agentic RAG over 1,240 legal PDFs, 93.6% precision, deterministic rules engine for fully auditable outputs. At Entrata: owned **20+ production modules end-to-end** (1M+ records, 100K+ MAU, 99.5% uptime), led Docker microservices rollout (50% speedup), AWS S3/RDS/Glue migration (100% PHI compliance), and first production AI pipeline on **1.6M real financial records**; built LIHTC (Tax Credit) and HUD Section 8/HCV compliance modules with deterministic rules engines and full audit trails. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Production AI · Compliance Automation · Platform Infrastructure · PropTech Domain

- Owned **20+ production modules end-to-end** (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — architecture, design patterns, rollout, adoption — **1M+ records, 100K+ MAU, 99.5% uptime**; optimized 200+ SQL queries (30% performance gain).
- Led **three platform-level initiatives**: Docker microservices rollout (50% execution speedup), AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents), and Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API) with quality gates, output validation, and HITL review before customer exposure.
- Built **LIHTC (Tax Credit) and HUD Section 8/HCV compliance modules**: AMI threshold calculations, income eligibility verification, annual/interim recertifications (HUD 50059), set-aside enforcement, and mixed-income portfolio tracking — deterministic rules engine eliminated **80% of manual workload** with full audit trails; zero compliance violations across 100K+ units.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Automation · API Integrations · Full-Stack Development

- Built income eligibility verification, AMI threshold enforcement, and household composition workflows processing **millions of records** with deterministic rules and full audit trails; built **API integrations and webhooks** across property management systems; mentored 5+ junior engineers; 300+ peer code reviews.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

AgentProbe: Multi-Agent AI Eval Framework
MULTI-AGENT LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling and inter-agent state handoffs; **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with HITL feedback loops; **CI/CD regression gates blocking failing runs before production**; 15-endpoint FastAPI backend; React observability dashboard — runs 100% locally at zero cost.

Leasel: Agentic Compliance Pipeline for Legal Documents
PROPTECH AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_raq

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs across 5 US states + federal standards — agentic LangGraph RAG pipeline: extraction, chunking, embedding, retrieval, deterministic rules engine for fully auditable high-stakes outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge and CI/CD regression gates.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R; Ruby (fast ramp — OOP/dynamic-typed foundation); FastAPI, Django, Flask, REST APIs, webhooks, RabbitMQ, SOA, event-driven distributed systems, microservices

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, Tool Calling, LLM-as-Judge, LLM Evaluation, AI Observability, HITL Feedback Loops, Prompt Engineering, Vector Stores (ChromaDB, pgvector), OpenAI API, Anthropic API, PyTorch, BERT, NLP, Embeddings, Responsible AI

Data & ML: Scikit-learn, XGBoost, Random Forest, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, Pandas, NumPy, Tableau, Power BI, ggplot2, Quarto, Jupyter

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, CI/CD, GitHub Actions, PostgreSQL, MySQL, Oracle, Git, Linux, HIPAA, PHI compliance, test-driven development, Agile/Scrum, scalable architecture, design patterns